

Afsana Bhuiyan

New York, NY | (347) 854-4495 | afsanab620@gmail.com | [linkedin.com/in/afsanabhuiyan/](https://www.linkedin.com/in/afsanabhuiyan/) | github.com/afsanab

EDUCATION

Rensselaer Polytechnic Institute

Bachelor of Science in Computer Science

Leadership: Discrete Mathematics Teaching Assistant, Girls Who Code Club President

August '20 – May '24

EXPERIENCE

Margin Research

Software Engineer (Contract)

New York, NY

February '26 – Present

- Took *Reagent* (SaaS code-analysis platform with graph-based threat detection in open-source software) from prototype to deployed v1 in 3 months for U.S. government customers, building its FastAPI service layer and React frontend from scratch.
- Designed and built a Celery-based orchestrator coordinating a five-stage analysis pipeline, using Redis/Valkey for task queuing and Elasticsearch to index and search graph analysis results.
- Built a reusable TypeScript graph-visualization tool that ingests NetworkX and JSON graphs, auto-detects node and relationship types, and surfaces per-node analytics including centrality and community detection; adopted across 2 internal projects.
- Containerized platform services, authoring Dockerfiles for deployment through the team's CI/CD pipeline, and wrote pytest suites covering the orchestrator's task routing and failure handling.

Pieces

Software Engineer Intern

New York, NY

September '25 – December '25

- Built a full-stack synthetic patient record generator (Flask REST API + JavaScript frontend) using the Gemini API to produce realistic EHR data for training and testing the company's AI document-summarization product.
- Designed the nested JSON schema for longitudinal patient records — demographics, conditions, and 4–6 encounters with vitals, labs, and separate nurse/provider note streams, validating clinical realism with two physician consultants.
- Engineered prompt specifications and a self-healing parsing pipeline that detects malformed LLM JSON output and auto-repairs it via a secondary repair prompt before re-validation, enabling reliable generation of 200+ clinician-note entries per patient.
- Implemented user-configurable batch generation and built export feature for datasets as structured JSON for downstream model training, with an interactive timeline UI featuring collapsible encounter views.

COOP Careers

Data Analytics Fellow

New York, NY

January '25 – June '25

- Built an interactive Tableau dashboard analyzing digital ad campaign performance (CPC, CPM, CPA, CTR, CVR) across U.S. states and device types, ranking top and bottom markets and flagging states that met CPA efficiency targets to guide ad-spend allocation.
- Cleaned and analyzed a 508-movie box office dataset using Python (pandas/NumPy), engineering profit metrics and Tableau visualizations that showed franchise presence and genre were the strongest drivers of profitability.
- Presented findings and strategic recommendations to a 500+ person audience.

NYC Board of Education

Substitute Teacher

New York, NY

September '24 – June '25

ISO New England

Software Engineer Intern

Holyoke, MA

June '23 – December '23

- Built an automated reporting pipeline (Java, Spring Boot, Apache Camel, Thymeleaf) that queries PL/SQL databases and emails the team scheduled dashboards of daily power-pricing data.
- Wrote a JUnit test suite for the Total Transfer Capability Calculator, a Java tool that models how much power can be transferred across New England's grid at projected future intervals, validating correctness of a domain-complex calculation engine.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, Java, SQL, C++
- **Frameworks:** React, Node.js, Flask, FastAPI, Spring Boot, Express
- **Infrastructure/Tools:** Celery, Elasticsearch, Redis/Valkey, Docker, Pytest, JUnit, Git, Linux, Tableau, Excel
- **Databases:** PostgreSQL, Supabase
- **Certifications:** Google Data Analytics

PROJECTS

LinkUP | Divhacks Hackathon

October '25

- Built the full React frontend for a transit-equity web app that calculates and compares commute times across multiple users to surface the fairest meetup location in NYC.
- Integrated Google Maps API for real-time map rendering and dynamic meetup-point visualization; coordinated with the backend team on API contracts to ensure live data updates reflected accurately across the UI.